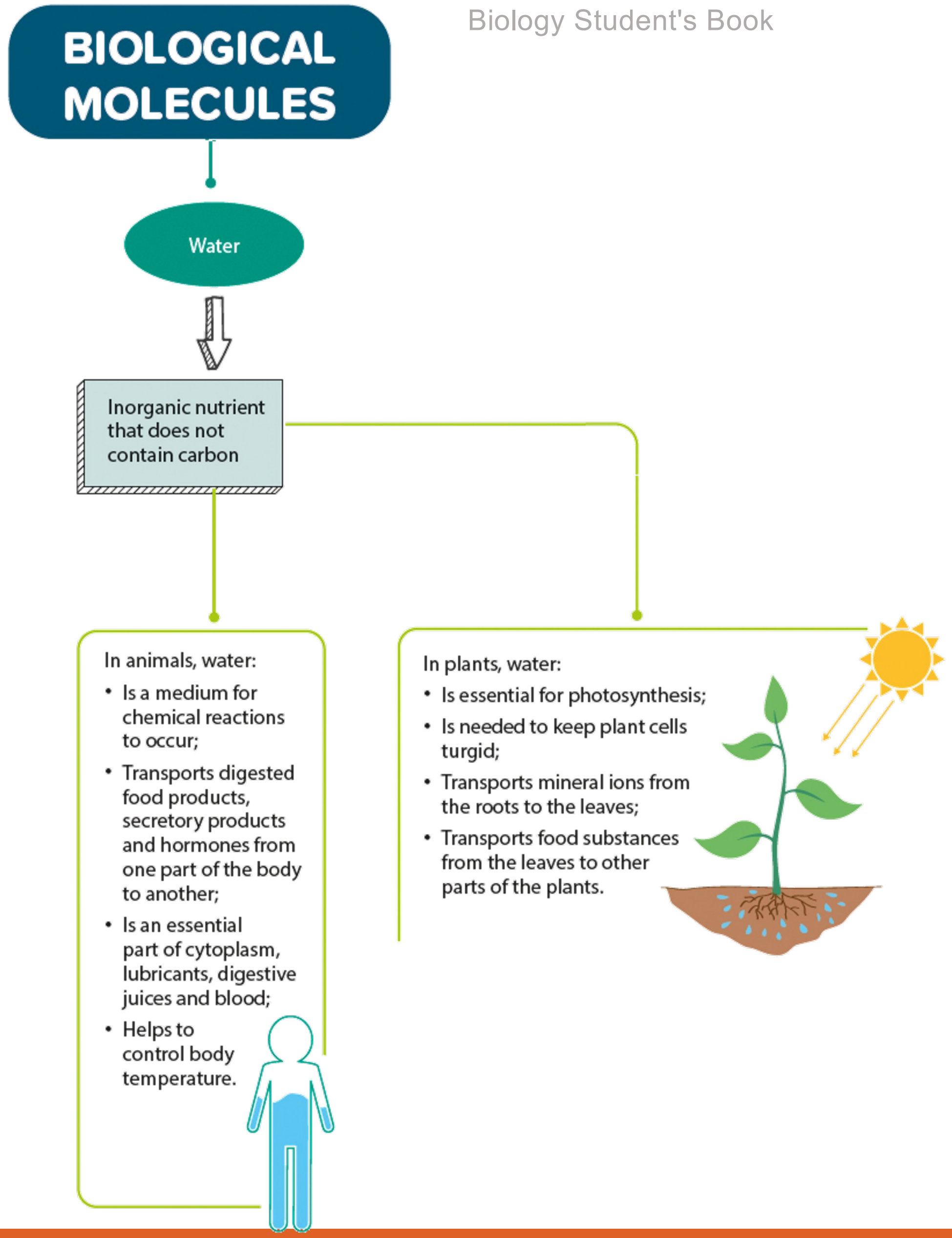
CHAPTER 4

BIOLOGICAL MOLECULES

4.1 The Need for Food

In this section, you will learn the following:

- State the principal dietary sources and describe the importance of water.



Why do we need food?

- Food provides energy for cell activities.
- Food substances are needed to make new cytoplasm.
- Food helps us stay healthy.

Why is water important?

- Water is an important solvent.
- Water helps in controlling body temperature.

- Water is needed for photosynthesis in plants.
- Water is involved in the transport of dissolved substances.

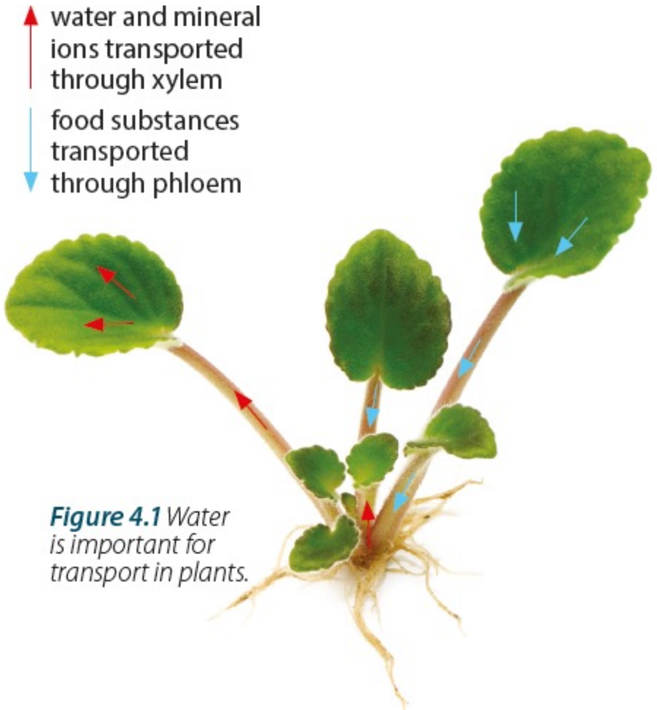


Figure 4.1 Water is important for transport in plants.

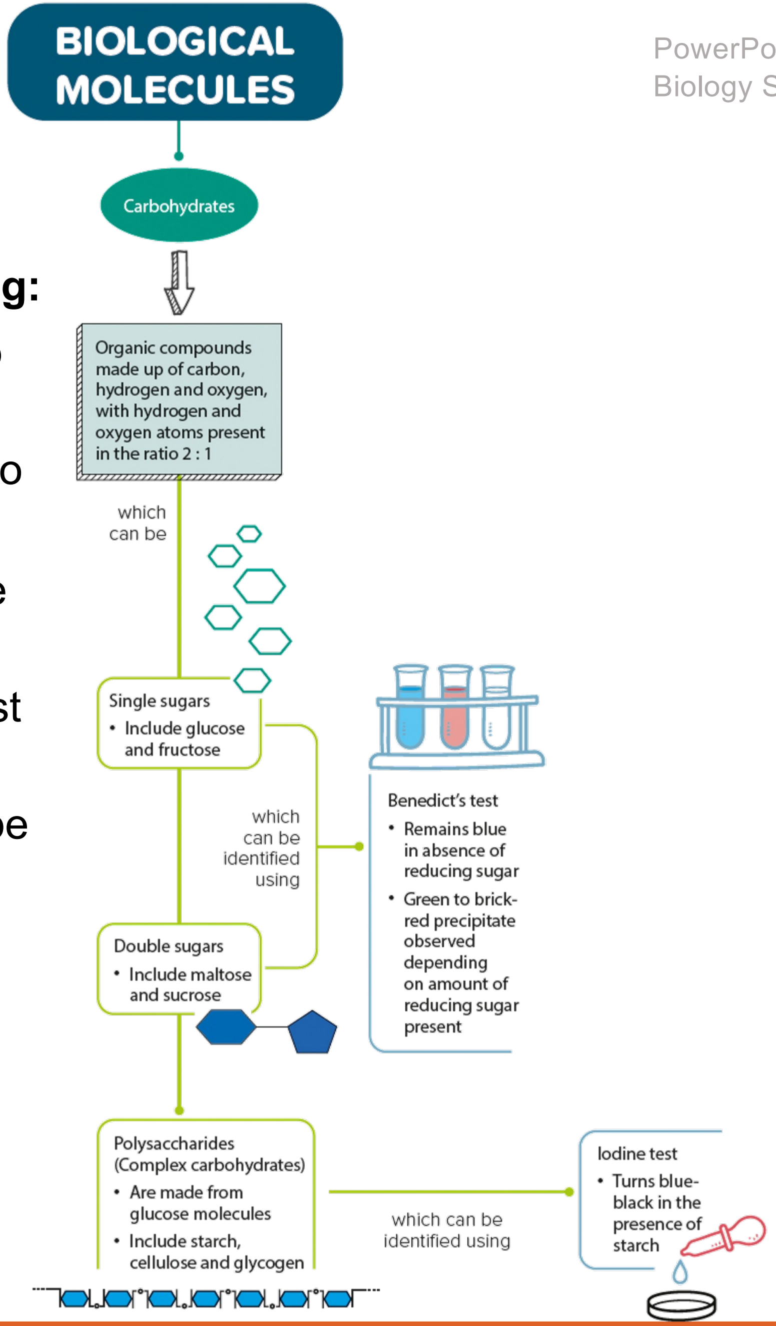
Let's Practise 4.1

1. Why is food important for us?
2. Discuss the roles of water in living organisms.

4.2 Carbohydrates

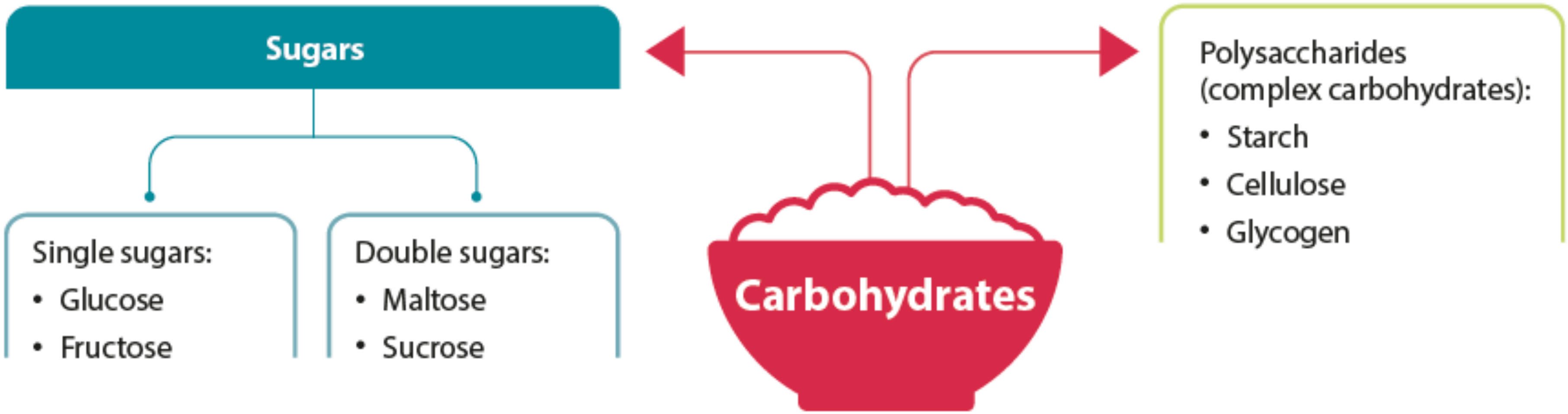
In this section, you will learn the following:

- List the chemical elements that make up carbohydrates.
- Describe the use of Benedict's solution to test for reducing sugars.
- State that starch, glycogen and cellulose are made from glucose.
- Describe the use of iodine solution to test for starch.
- State the main food sources and describe the importance of carbohydrates.



What are carbohydrates?

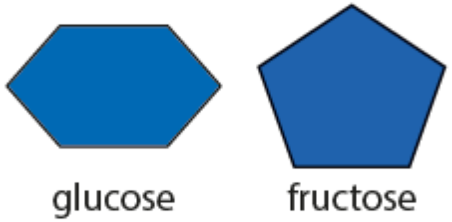
- **Carbohydrates** are organic molecules made up of the elements carbon, hydrogen and oxygen.
- The hydrogen and oxygen atoms are present in the ratio 2 : 1.



Classification of carbohydrates

What are single sugars?

- Single sugars cannot be further digested into smaller molecules.
- They can pass through cell membranes and be absorbed into the cells.
- Glucose and fructose are common single sugars.

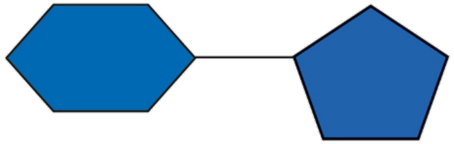


What are double sugars?

- A double sugar is made up of two single sugars bonded together.
- Maltose is produced from two glucose molecules.

Reaction between two glucose molecules to give one maltose molecule and water

- Sucrose is made up of one glucose molecule and one fructose molecule joined together.



Sucrose

Sucrose can be found in sugarcane stems, sweet fruits and certain storage roots such as carrots.

LINK

Sucrose is the sugar that is transported by the phloem in plants. You will learn more about this in Chapter 8.

How do we test for reducing sugars?

Let's Investigate 4A

- **Benedict's solution** can be used to test for the presence of reducing sugars such as glucose, fructose, galactose, maltose and lactose.
- The table shows the colour changes observed when carrying out the Benedict's test with increasing amounts of reducing sugar.

Observation	Amount of reducing sugar present
The solution remains blue.	No reducing sugar
A blue to green precipitate is formed.	Traces of reducing sugar
A blue to yellow or orange precipitate is formed.	Moderate amount of reducing sugar
A blue to brick-red precipitate is formed.	Large amount of reducing sugar

What are complex carbohydrates?

- A complex carbohydrate or **polysaccharide** (Greek: *poly* = many) consists of many monosaccharide molecules joined together.
- **Starch**, **glycogen** and **cellulose** are complex carbohydrates made up of glucose molecules.

Examples of food rich in starch

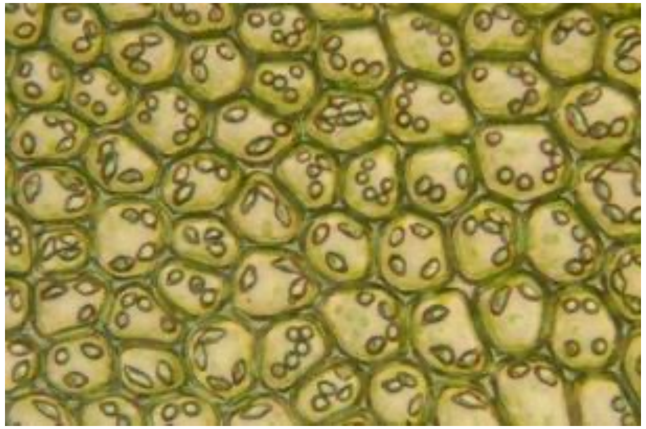
How do we test for starch?

Let's Investigate 4B

- Starch can be detected by the **iodine test**.
- Iodine solution turns blue-black upon contact with the potato.

What are the functions of carbohydrates?

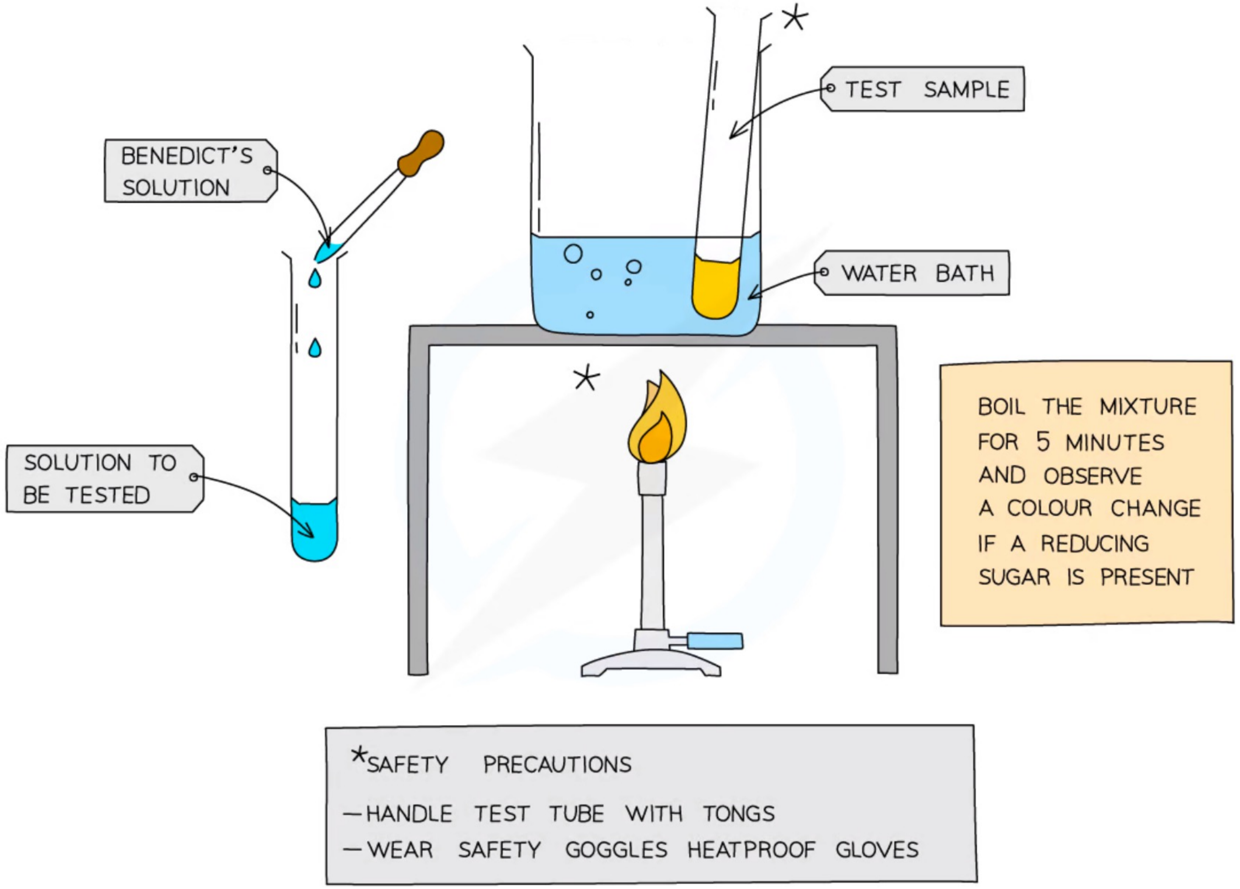
- As a source and store of energy
- Form supporting structures, such as cellulose cell walls in plants



- For the synthesis of nucleic acids
- To make lubricants, such as mucus which covers the inner surface of the trachea and traps dust particles
- To make nectar in flowers

Test for glucose (a reducing sugar)

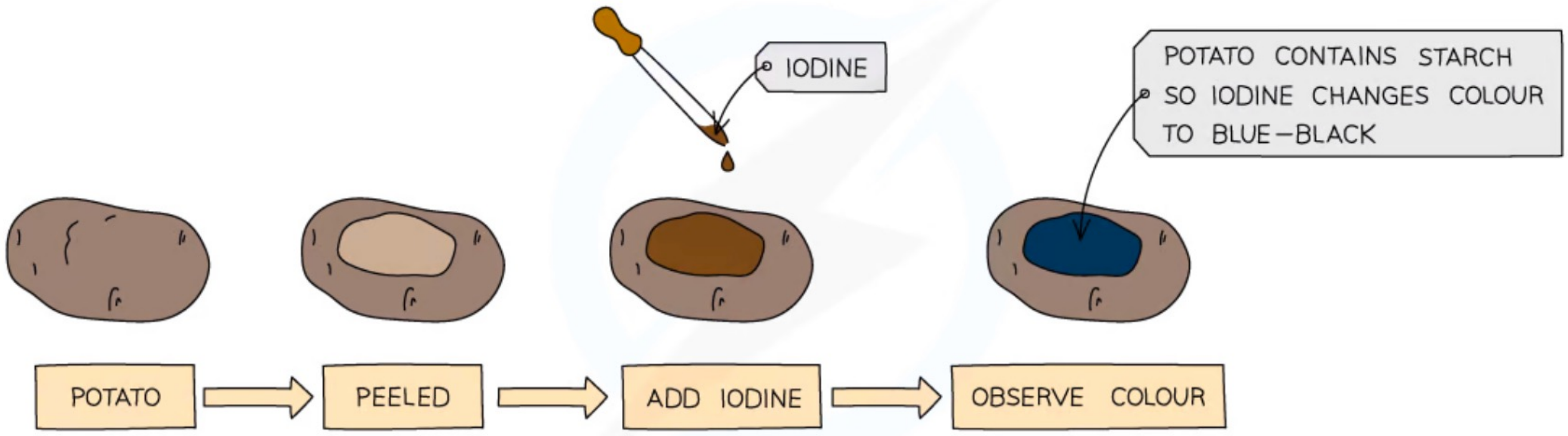
- Add **Benedict's solution** into sample solution in test tube
- **Heat** at 60 – 70 °c in water bath for **5 minutes**
- Take test tube out of water bath and observe the colour
- A positive test will show a colour change from **blue to orange or brick red**



Test for starch using iodine

We can use iodine to test for the presence or absence of starch in a food sample.

- Add drops of **iodine solution** to the food sample
- A positive test will show a colour change from **orange-brown to blue-black**



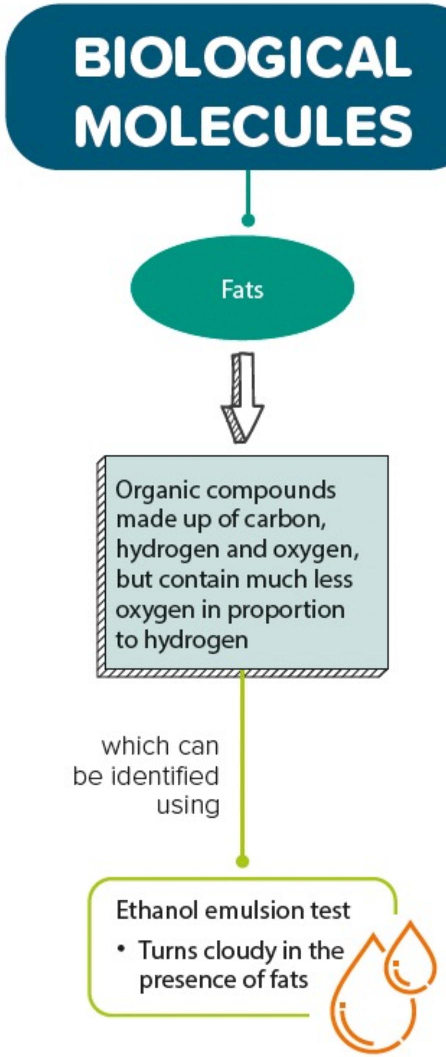
Let's Practise 4.2

1. How can you identify reducing sugars?
2. How can you identify starch?
3. Outline the role of carbohydrates in the life of a living organism.

4.3 Fats

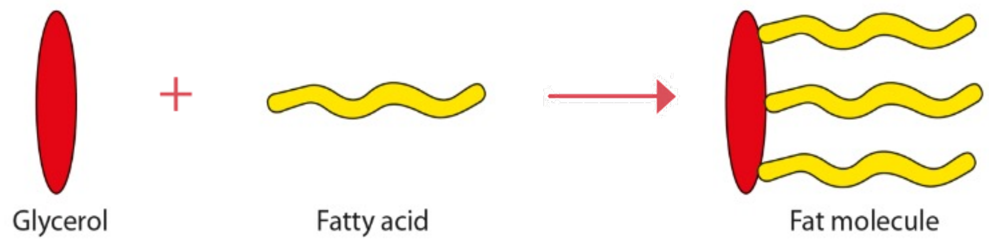
In this section, you will learn the following:

- List the chemical elements that make up fats.
- State that fats and oils are made from fatty acids and glycerol.
- Describe the use of the ethanol emulsion test to test for fats and oils.
- State the main food sources and describe the importance of fats and oils.



What are fats?

- **Fats** are organic molecules made up of carbon, hydrogen and oxygen.
- Unlike carbohydrates, fats contain much less oxygen in proportion to hydrogen.
- A fat molecule is made up of one molecule of glycerol and three molecules of fatty acids.
- Foods that are rich in fats include butter, cheese, fatty meat, olives, many nuts, peas, beans, seeds or castor oil and palm oil.



Examples of food rich in fats

HELPFUL NOTES

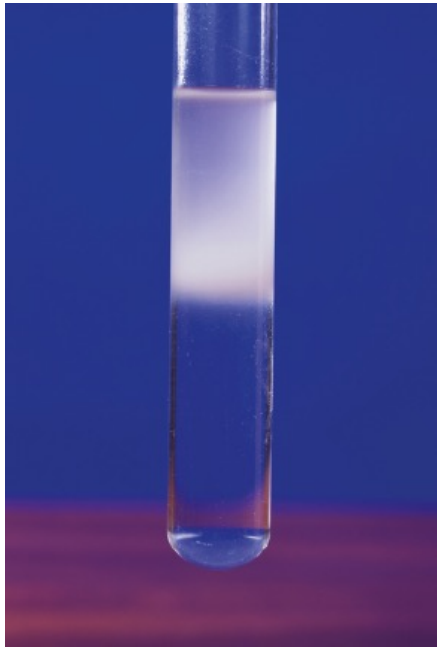
Lipids are compounds that are soluble in organic solvents such as chloroform and ethanol. Examples of lipids include fats, waxes and steroids.

Is there any difference between fats and oil? Fats are solids and oils are liquids at room temperature. In this book, we shall use the term *fats* to refer to both solid and liquid fats.

How do we test for fats?

Let's Investigate 4C

- The **ethanol emulsion test** is a test for the presence of fats.
- A cloudy white emulsion is formed when water is added to the mixture of ethanol and coconut oil, which indicates the presence of fats in coconut oil.
- A cloudy white emulsion is also formed when water is added to ethanol that has been mixed with peanuts, which indicates the presence of fats in peanuts.



What are the functions of fats?

- As a source and storage of energy
- As an insulating material that prevents excessive heat loss

Seals have a layer of fat beneath the skin to reduce body heat loss.

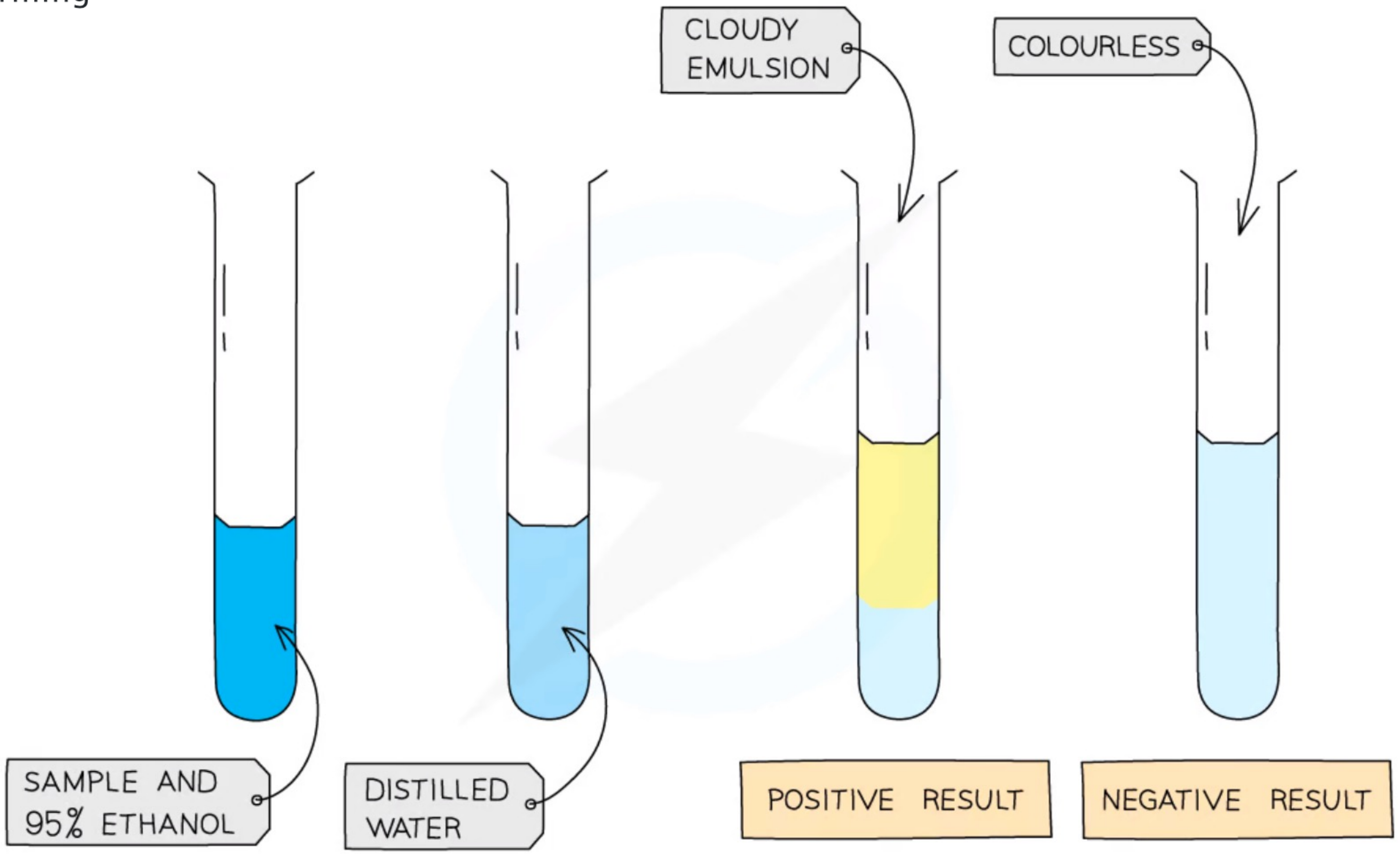
- As a solvent for fat-soluble vitamins and hormones
- As an essential part of cell membranes
- Reduce water loss from the skin surface

Let's Practise 4.3

1. List the chemical elements that make up fats.
2. How can you identify fats?

Test for lipids

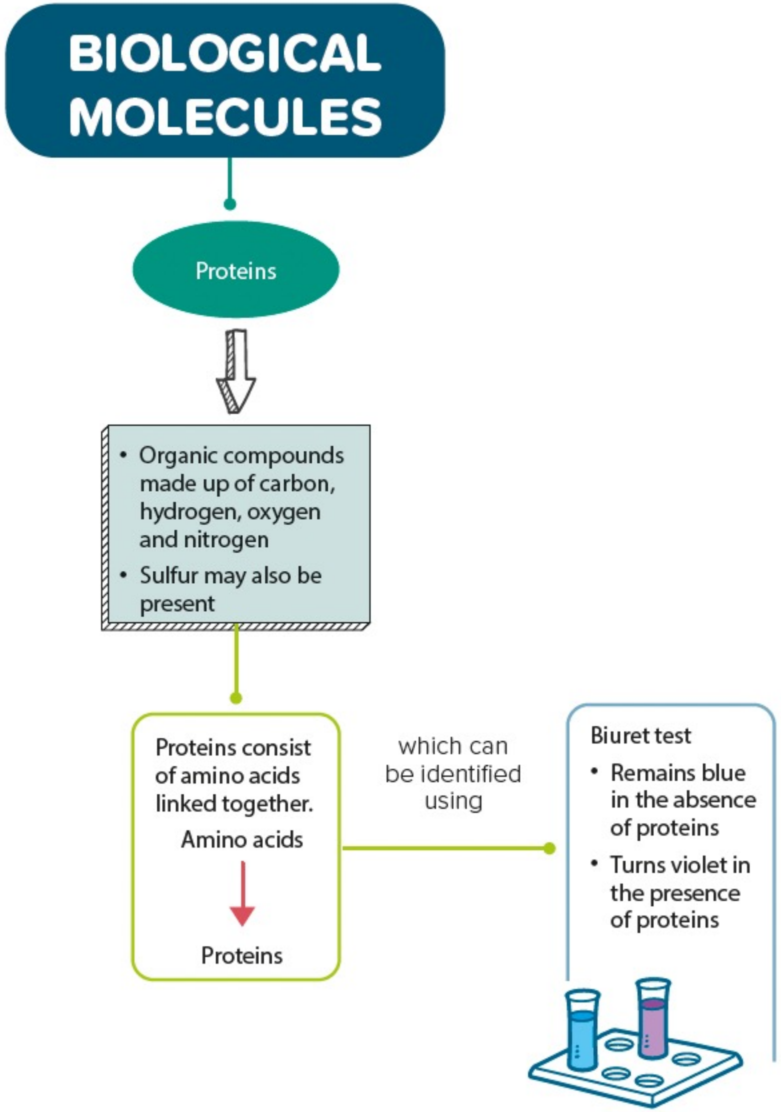
- Food sample is mixed with **2cm^3 of ethanol** and shaken
- The ethanol is added to an equal volume of **cold water**
- A positive test will show a **cloudy emulsion** forming



4.4 Proteins

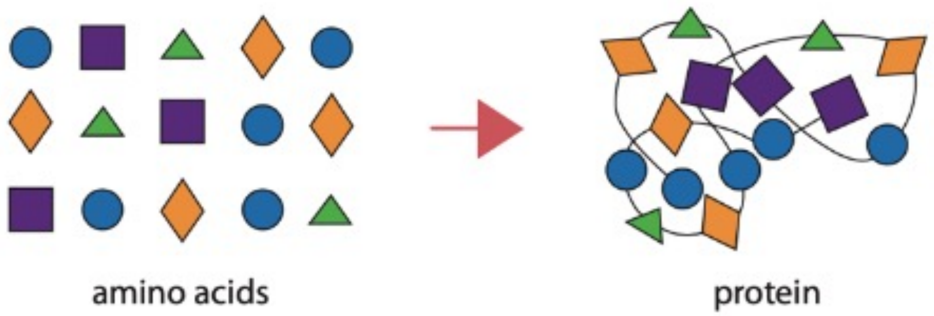
In this section, you will learn the following:

- List the chemical elements that make up proteins.
- State that proteins are made from amino acids.
- Describe the use of Biuret solution to test for proteins.
- State the main food sources and describe the importance of proteins.



What are proteins?

- **Proteins** are organic molecules made up of carbon, hydrogen, oxygen and nitrogen. Sulfur may also be present.
- Proteins are present in all cells.
- A protein molecule is built up from simpler compounds known as amino acids.



Amino acids join to form a polypeptide chain that folds into a protein.

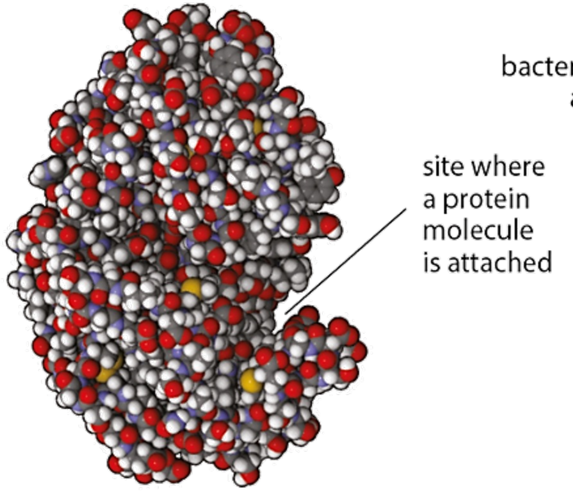
HELPFUL NOTES

What is the difference between a protein and a polypeptide?
A polypeptide is like a long thread, whereas a protein is like the sweater that is knitted out of that thread. Can you see that the knitted thread (sweater) has a "shape"?

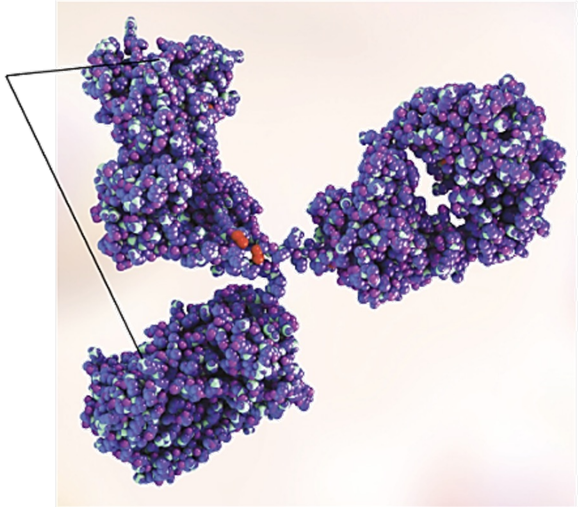
What are proteins?

- Enzymes and antibodies are made of proteins.
- Each protein has a unique three-dimensional shape that enables it to perform its function.
- Foods rich in proteins include milk, eggs, seafood, meat, soya beans, nuts, grains and vegetables such as French beans.

Examples of food rich in proteins



▲ Pepsin (an enzyme) has a specific site to attach to a protein and break it down during digestion.



▲ An antibody has specific sites to attach to bacteria or viruses and destroy them.

How can we test for proteins?

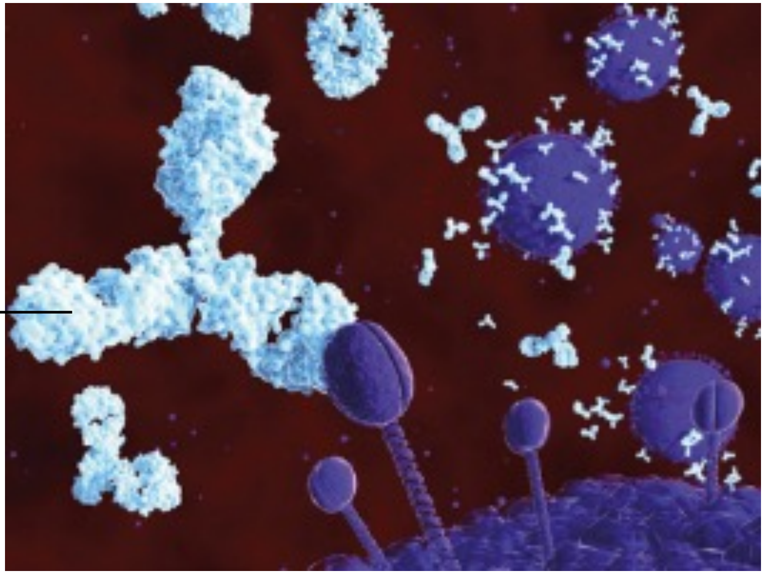
Let's Investigate 4D

- Proteins can be detected using the **biuret reagent**.
- A violet (deep purple) colour is observed when the egg white solution is mixed with biuret reagent. This indicates the presence of proteins.
- In the absence of proteins, the solution will remain blue.

What are the functions of proteins?

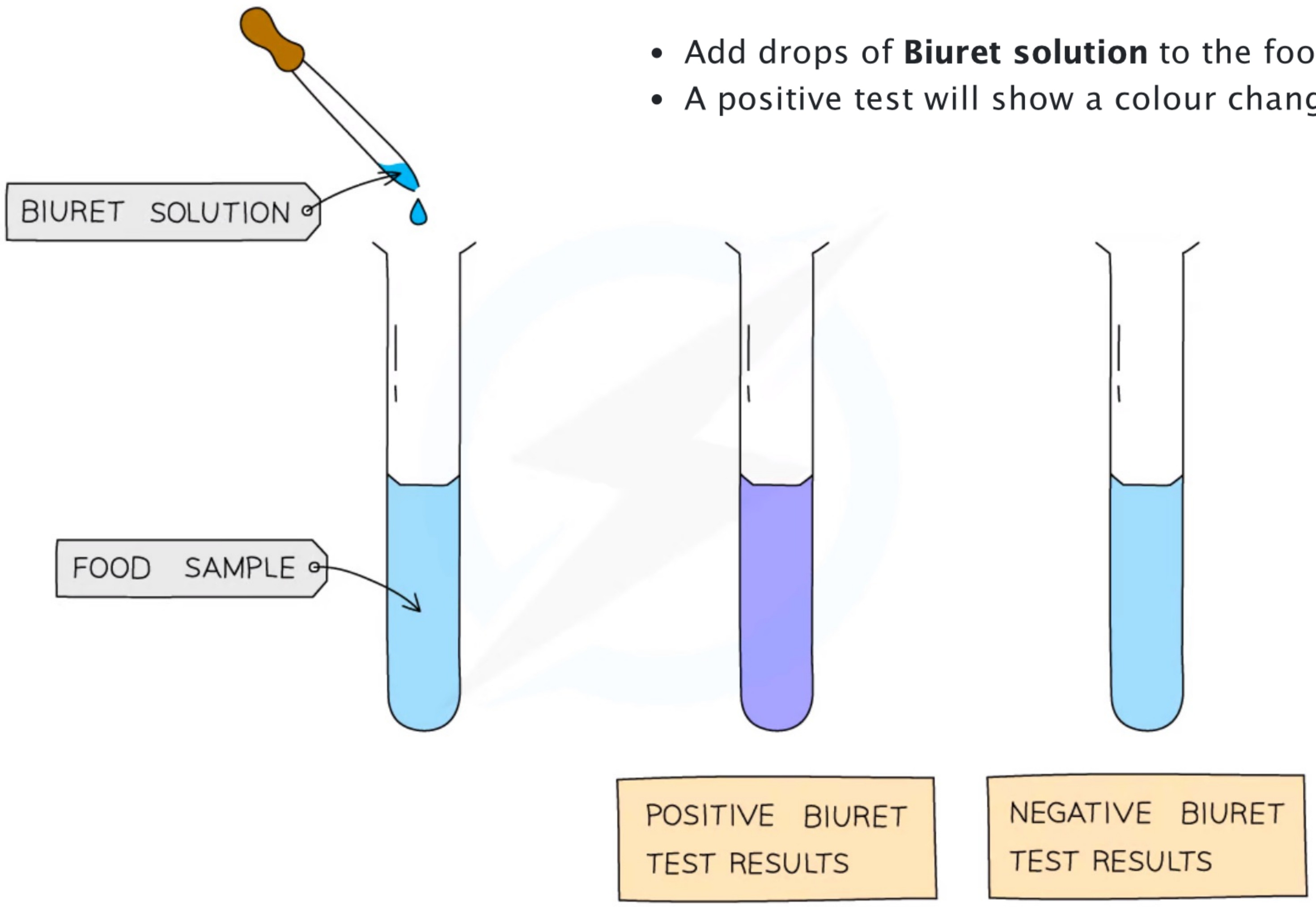
- To make new cytoplasm in cells
- To make enzymes and some hormones
- For the formation of antibodies to combat diseases
- To make chromatin in chromosomes

antibody



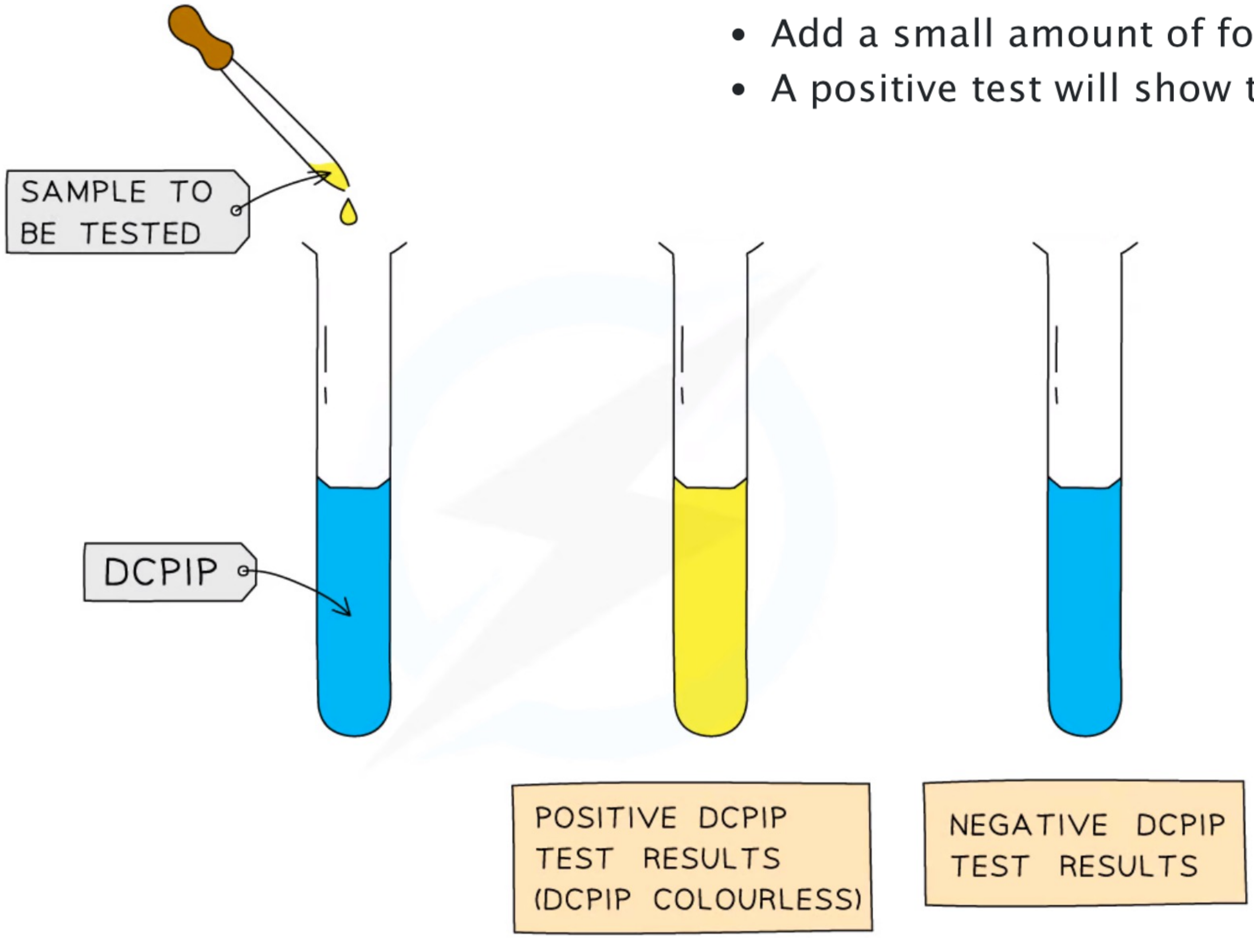
Test for protein

- Add drops of **Biuret solution** to the food sample
- A positive test will show a colour change from **blue to violet / purple**



Test for vitamin C

- Add **$1\,cm^3$ of DCPIP solution** to a test tube
- Add a small amount of food sample (as a solution)
- A positive test will show the **blue colour of the dye disappearing**



Let's Practise 4.4

1. List the chemical elements that make up proteins.
2. How can you identify proteins?
3. (a) Food tests were carried out on three different types of food, labelled A, B and C. The results of the tests are given in the table. Using the results from the table, state what substances are present in each food.

Test	A	B	C
Add iodine solution.	Iodine remained brown.	Iodine remained brown.	A blue–black colour was observed.
Heat with Benedict's solution.	A brick-red precipitate was formed.	A brick-red precipitate was formed.	The solution remained blue.
Add biuret reagent.	The solution remained blue.	A violet colour was observed.	A violet colour was observed.

(b) Describe how you would test a piece of skin from an orange fruit for the presence of fats.

(c) Outline the role of proteins in the life of a living organism.

MOLECULE	CHEMICAL ELEMENTS
CARBOHYDRATE	CARBON, OXYGEN AND HYDROGEN
PROTEIN	ALL CONTAIN CARBON, OXYGEN, HYDROGEN AND NITROGEN AND SOME CONTAIN SMALL AMOUNTS OF OTHER ELEMENTS SUCH AS SULPHUR
LIPID	CARBON, OXYGEN AND HYDROGEN

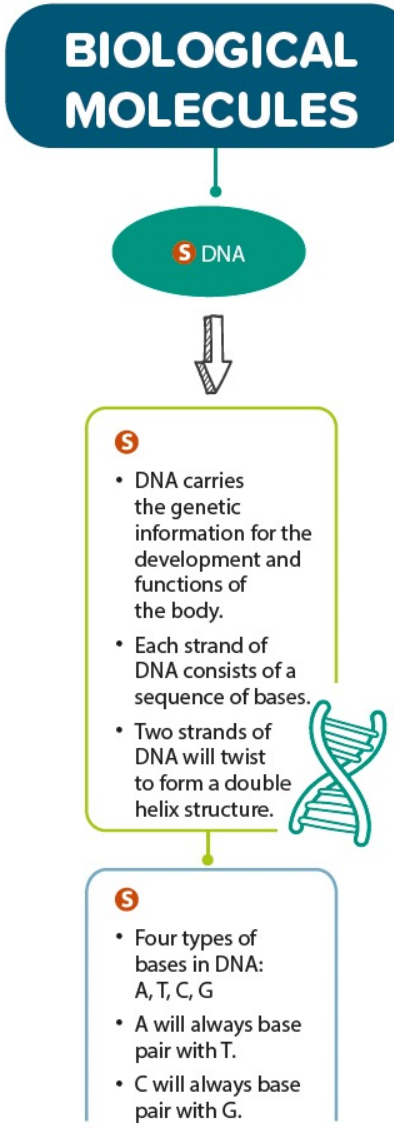
Exam Tip

When describing food tests in exam answers, make sure you give the **starting colour** of the solution and **the colour it changes to** for a positive result.

4.5 DNA (Deoxyribonucleic Acid)

In this section, you will learn the following:

- Describe the structure of DNA.
- Explain the use of the sequences of bases in DNA in classification.
- Explain that groups of organisms share a more recent ancestor and are more closely related if they have sequences of bases in DNA that are more similar.

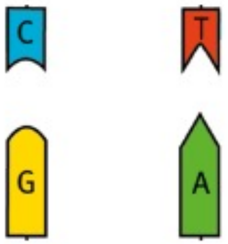


What is DNA?

- **DNA (deoxyribonucleic acid)** carries the genes or genetic information for the development and functions of the body.
- A DNA molecule has a double helix structure.

Part of a DNA molecule

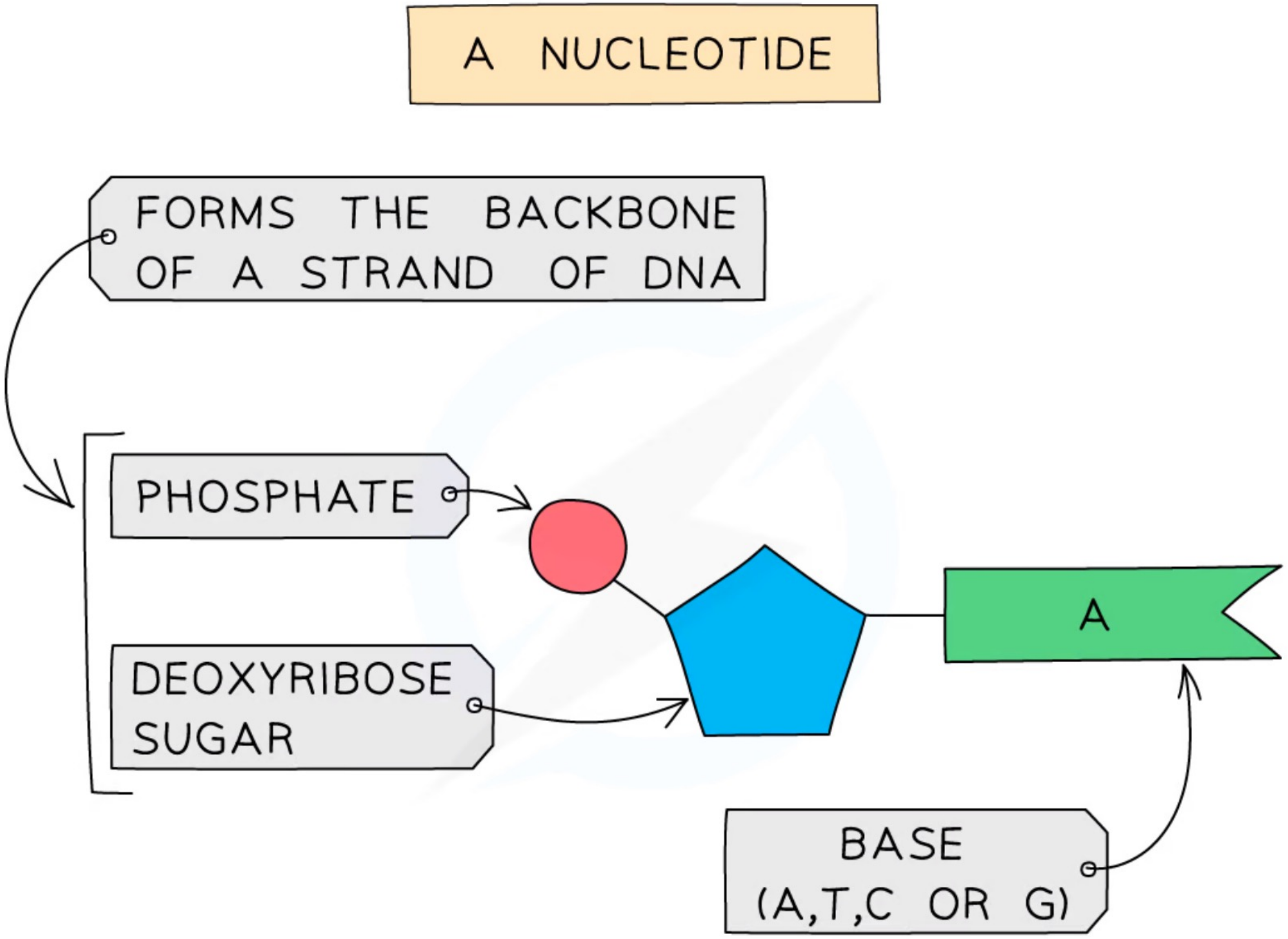
- There are four types of bases in DNA — A, C, G and T.
- A will always pair with T, and C will always pair with G.



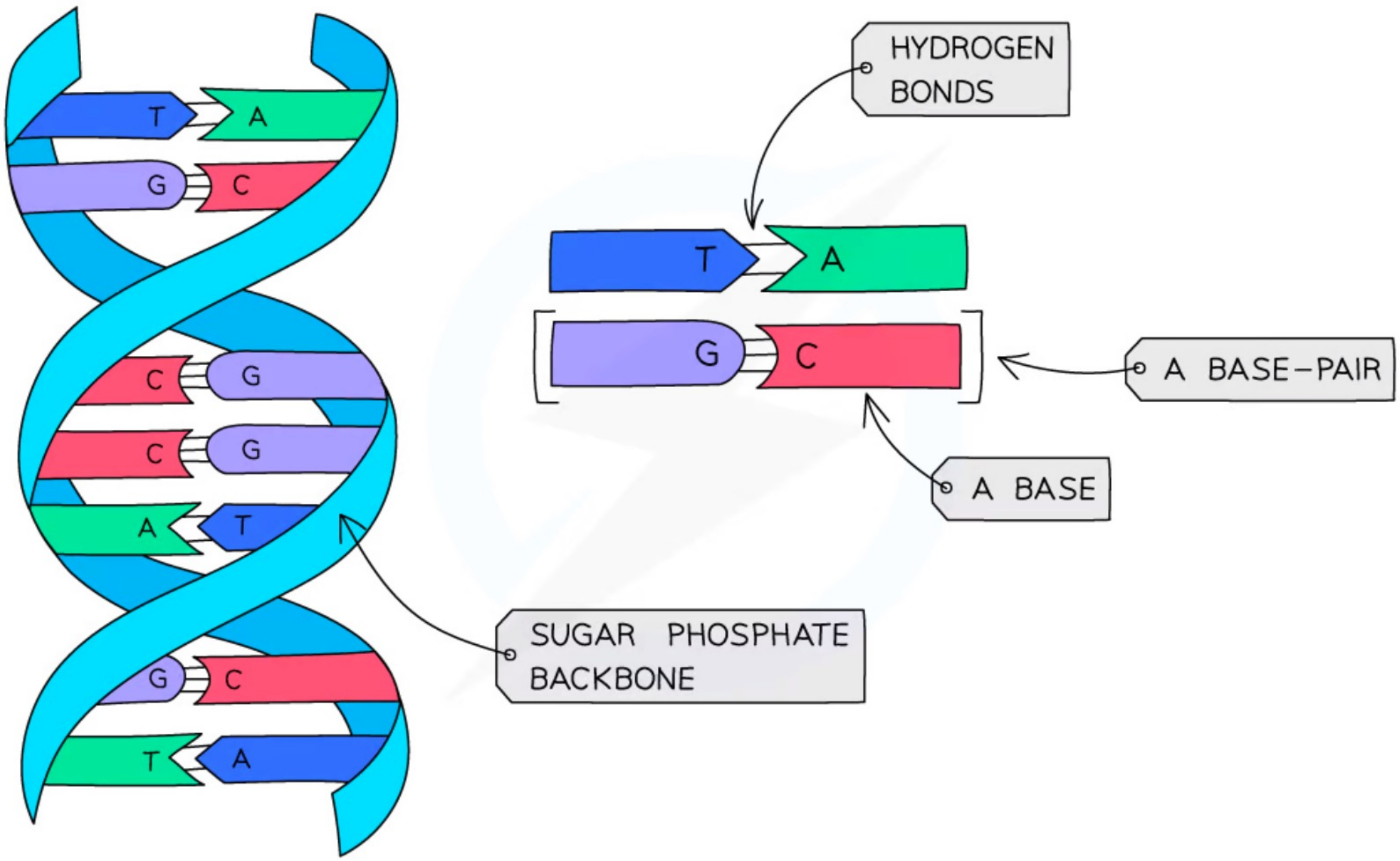
Base pairing

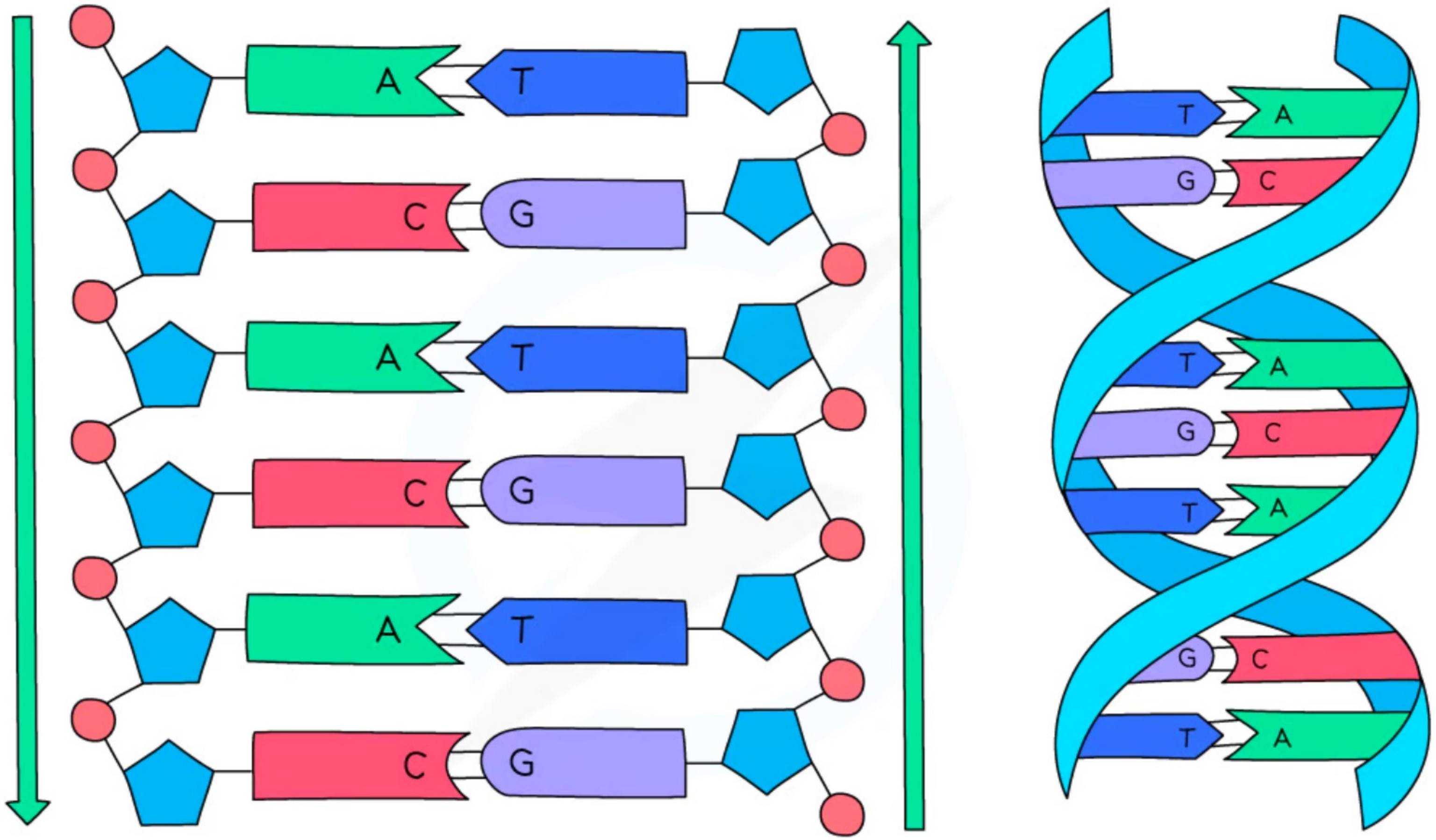
HELPFUL NOTES

The DNA molecule is like a twisted ladder. The bases are joined up to form the "rungs" of the ladder. Each "rung" consists of a pair of bases.


A NUCLEOTIDE
FORMS THE BACKBONE OF A STRAND OF DNA
PHOSPHATE
DEOXYRIBOSE SUGAR
A
BASE (A,T,C OR G)

- All nucleotides contain the same phosphate and deoxyribose sugar, but differ from each other in the **base** attached
- There are four different bases, **Adenine (A), Cytosine (C), Thymine (T) and Guanine (G)**
- The bases on each strand pair up with each other, holding the two strands of DNA in the double helix
- The bases always pair up in the same way:
    - Adenine always pairs with Thymine **(A–T)**
    - Cytosine always pairs with Guanine **(C–G)**

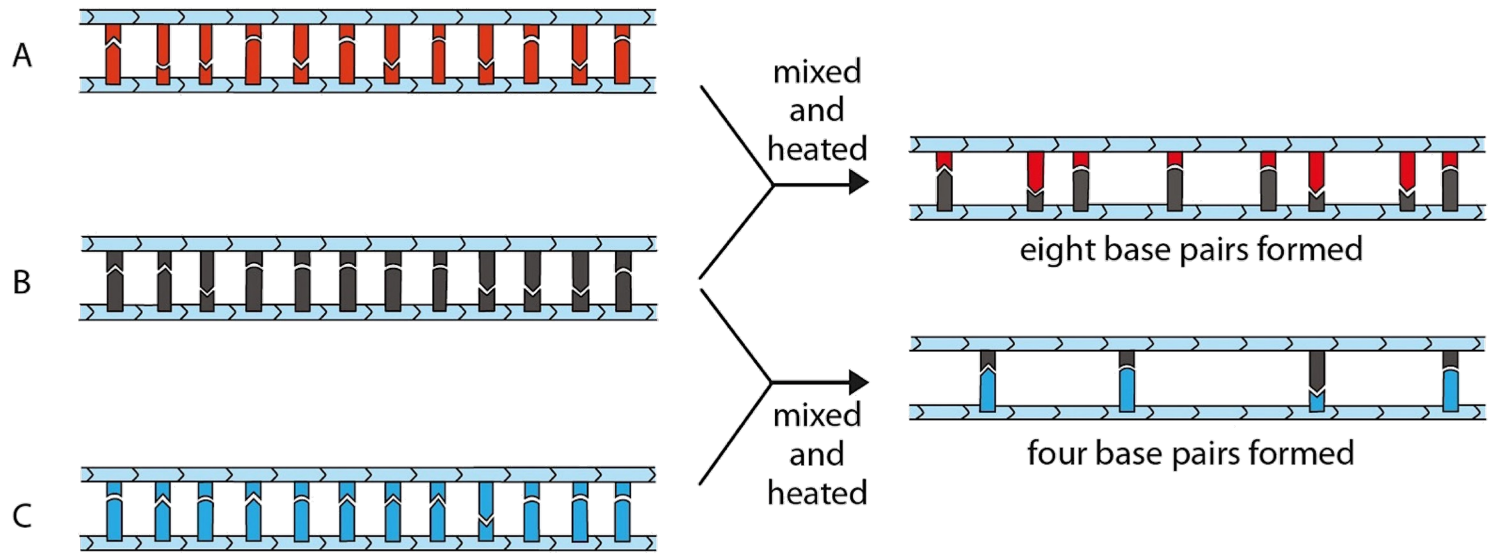

T
A
G
C
C
G
C
G
A
T
G
C
T
A
HYDROGEN BONDS
T
A
G
C
A BASE-PAIR
A BASE
SUGAR PHOSPHATE BACKBONE



EACH STRAND IS USED SEPARATELY, A GENE IS A SEQUENCE OF BASES – NOT A SEQUENCE OF BASE PAIRS!

What is DNA?

- Each species of organism has a unique sequence of bases in DNA.
- We can determine how closely related two species of organisms are by comparing their DNA.



Using DNA to determine how closely different organisms are related (Figure 4.29 of Student's Book)

There are more base pairs between organisms A and B than between B and C. This means that A and B have more similar base sequences. They are more closely related to each other.

Exam Tip

You do not need to learn the names of the bases, **just their letter**. Make sure you **know which bonds with which**, as this is the most commonly asked question about this topic.

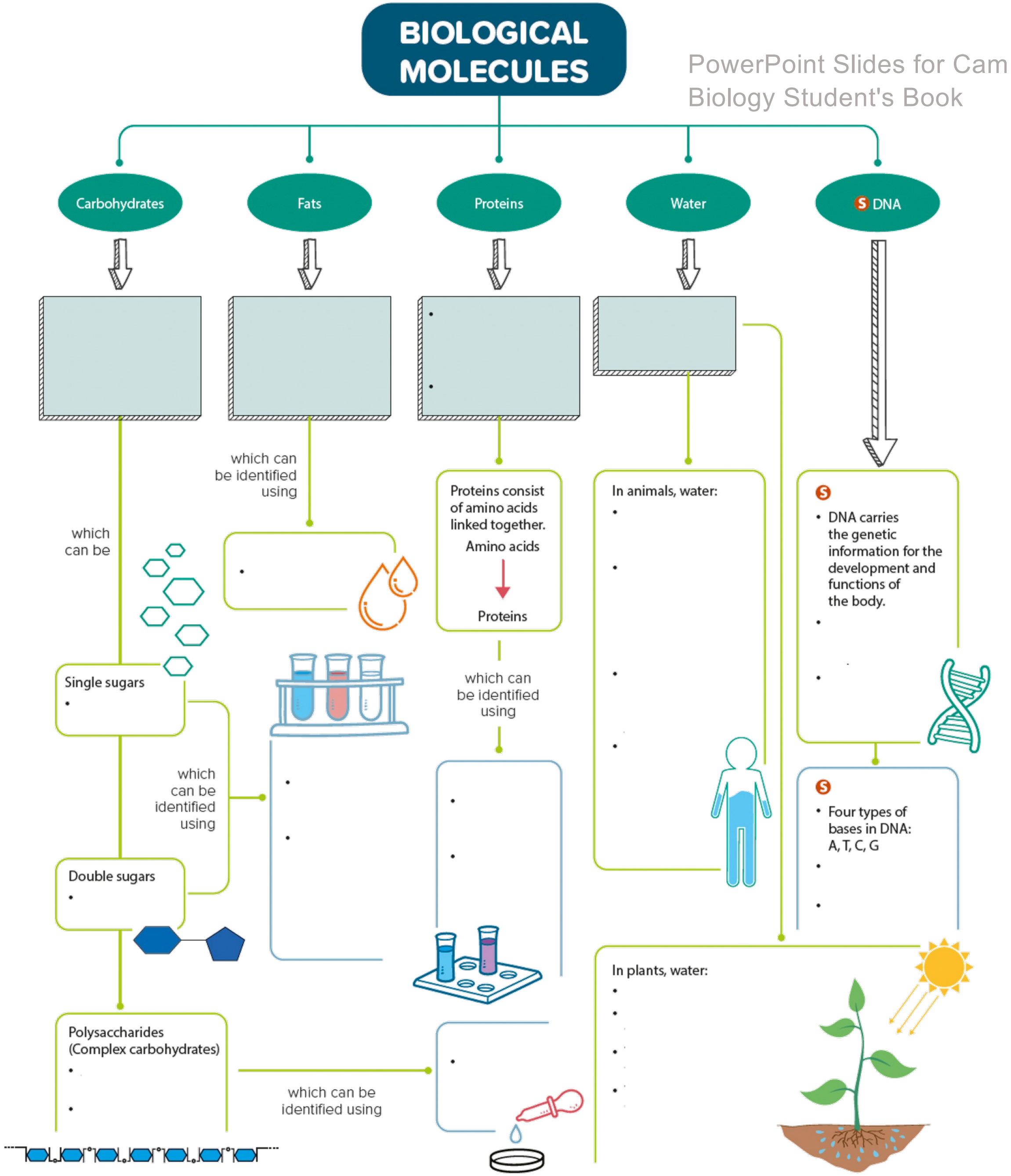
Let's Practise 4.5

1. What does DNA stand for?
2. Name the bases that make up the DNA molecule.

Marshall Cavendish Education

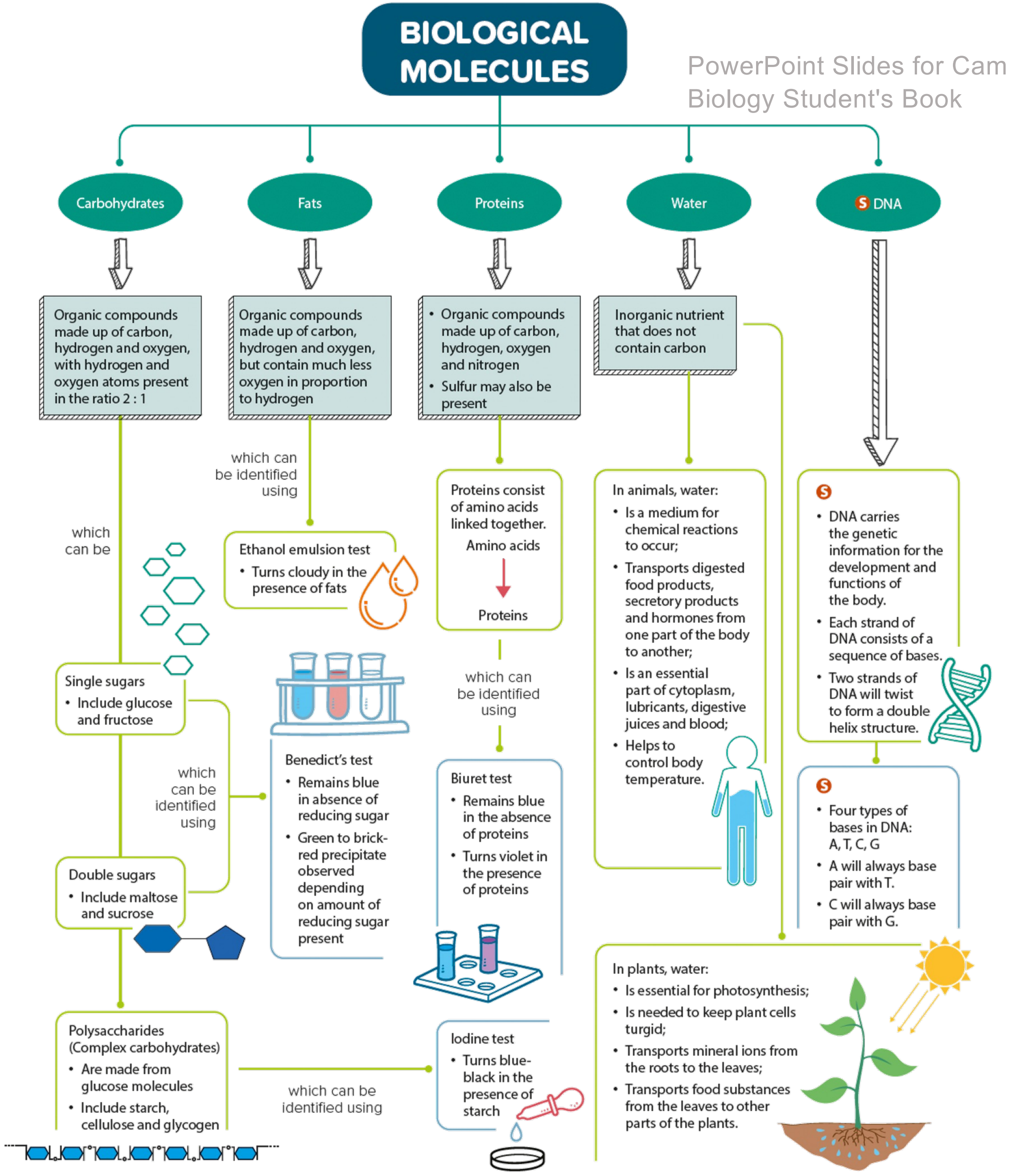
What have you learnt?

Can you draw your own mind map?



What have you learnt?

Can you draw your own mind map?



Percent Change

$$\text{Percent Change} = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}} \times 100\%$$

If the result is positive, it is an increase.
If the result is negative, it is a decrease.